THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A person has no pancreas. Which will be the most affected? 1

(a) absorption of oxygen **(b)** digestion of all three food types
(c) breathing **(d)** digestion of water only

2. A variegated leaf kept in sunlight: only the green part turns blue-black. This shows: 1

(a) chlorophyll is needed **(b)** water is not needed
(c) light is not needed **(d)** CO2 is not needed

3. If the angle of incidence of a ray on a plane mirror is increased by 15°, the angle of reflection: 1

(a) decreases by 15° **(b)** doubles **(c)** stays the same **(d)** increases by 15°

4. If the Earth rotated twice as fast, the day would be about: 1

(a) 48 hours **(b)** 6 hours **(c)** 12 hours **(d)** 24 hours

5. In a neutralisation, if 15 mL of acid is needed for 25 mL of a base, then for 50 mL of the base it will be: 1

(a) 25 mL **(b)** 15 mL **(c)** 60 mL **(d)** 30 mL

6. A tester lights when connected across tap water but only dimly across distilled water. Reason: 1

(a) tap water has dissolved salts **(b)** distilled water is wet
(c) tap water is pure **(d)** distilled water has salts

7. In $\mathrm{Fe} + \mathrm{CuSO_4} \longrightarrow \mathrm{FeSO_4} + \mathrm{Cu}$, the metal that displaces copper is: 1

(a) Cu **(b)** $\mathrm{CuSO_4}$ **(c)** Fe **(d)** $\mathrm{FeSO_4}$

8. Which of the following would you use to show that a new gas is formed? 1

(a) lime water test **(b)** balance **(c)** magnet **(d)** ruler

9. If the ovaries do not function, a girl may not have: 1

(a) digestion **(b)** breathing **(c)** growth **(d)** menstruation

10. At the junction between the two breezes at around dusk, the wind: 1

(a) nearly stops **(b)** becomes a storm
(c) doubles **(d)** reverses at once strongly

11. A train leaves at 6 a.m. and runs at a steady 60 km/h. By 12 noon it has covered: 1

(a) 720 km **(b)** 60 km **(c)** 360 km **(d)** 120 km

12. A person breathes 16 times a minute, taking in 500 mL each time. The volume of air in one minute is: 1

(a) 16 mL **(b)** 8000 mL **(c)** 4000 mL **(d)** 500 mL

13. Which energy change occurs in respiration? 1

(a) chemical energy to usable energy **(b)** sound to chemical
(c) heat to chemical **(d)** light to chemical

14. During a lunar eclipse, the Moon passes into Earth's: 1

(a) shadow **(b)** light **(c)** orbit only **(d)** atmosphere only

15. The Moon takes 27.3 days to complete a revolution but 29.5 days from full moon to full moon because: 1

(a) the Earth moves around the Sun meanwhile **(b)** the Moon stops
(c) the Sun moves **(d)** the Moon slows down

16. Which statement about milk of magnesia is correct? 1

(a) it is a neutral salt used as a drink **(b)** it is an acid used to relieve acidity
(c) it is a base used to relieve acidity **(d)** it is an indicator

17. **Assertion (A):** Photosynthesis takes place in the presence of sunlight and chlorophyll. 1
Reason (R): Chlorophyll absorbs the energy of sunlight needed to make food.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** A shadow is formed when an opaque object comes in the path of light. 1
Reason (R): Light travels in straight lines.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** Day and night occur because of the rotation of the Earth. 1
Reason (R): The Earth rotates on its axis once in about 24 hours.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** Red litmus paper turns blue when dipped in lemon juice. 1
Reason (R): Lemon juice is acidic in nature.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Why are the pins of a plug made of brass or copper? 2

22. Why do non-metallic oxides cause acid rain? 2

23. Explain why the burning of paper is a chemical change but its tearing is a physical change. 2

24. Why do boys have a higher proportion of muscle growth during adolescence? 2

25. Why do we place a cloth wet with water around a matka in summer? 2

26. Why can't the average speed of a round trip be the average of two speeds? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Explain why two spoons, one of steel and one of plastic, in a cup of boiling water show different results. 3

28. A bus leaves a town at 5 a.m. and runs at 40 km/h until 8 a.m. Find the distance covered. 3

29. Explain why white blood cells are found in higher numbers at an infected wound. 3

30. Why do trees in a dense forest have most of their leaves at the top? 3

31. Describe how to use a stick and its shadow to find the time of noon. 3

OR

Explain how a total lunar eclipse might last over an hour but a total solar eclipse only minutes. 3

32. Write the neutralisation equation for the formation of potassium sulphate. 3

OR

Describe the safe way of fixing a bulb in a holder. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Rahul, aged 13, notices that he is growing taller quickly, hair is appearing on his face, and his voice sometimes 'cracks' while speaking. He feels shy about these changes, but his teacher explains that they are normal. 5

(i) What is the stage of life in which these changes occur called?
(ii) Name the male sex hormone responsible for them.
(iii) Name the organ that secretes this hormone.
(iv) Explain why Rahul's voice becomes deeper.

34. **Case study.** Sunita is cooking dal on a gas stove. She leaves a steel spoon in the pan and, after some time, the spoon's handle becomes too hot to hold. She uses a wooden spoon to stir. She also sees the dal circulating inside the pan as it boils. 5

(i) By which mode does the handle of the steel spoon get hot?
(ii) Why is the wooden spoon preferred?
(iii) By which mode does the dal circulate?
(iv) Explain how the water in a pan gets heated from the bottom to the top.

35. Compare a pinhole camera and a plane mirror as image forming devices. 5

36. Explain how astronomers predict the time of eclipses using the motions of the Sun, Earth and Moon. 5

OR

Explain, with the help of examples, how the ideas of acids, bases and neutralisation connect agriculture, health and the environment. 5

37. Plan an energy-saving programme for a home with five measures and say why each helps. 5

OR

Explain, with equations, the reactions of magnesium with oxygen, acid and copper sulphate solution. 5

38. Analyse the changes taking place when you light a gas stove under a vessel of milk, with milk boiling over and burning at the bottom. Classify each change. 5

OR

Compare the roles of nervous system and endocrine system in adolescent changes and in responding to stress. 5

END OF QUESTION PAPER **Total: 80 Marks**

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